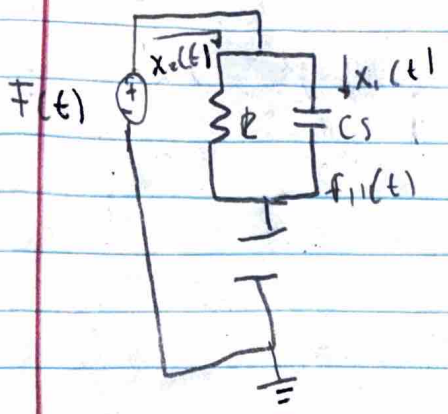
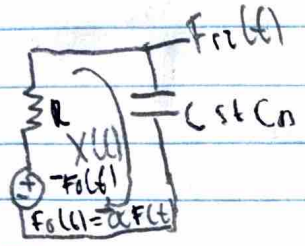




Analisis del circuito por superposicion

circuito 1. Apagando $F_0(t)$

$$X_0(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{11}(t)}{dt}$$

$$X_1(t) = C_2 \frac{d[F(t) - F_{11}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{11}(t)}{R}$$

$$C_p \frac{dF_{11}(t)}{dt} = C_2 \frac{d[F(t) - F_{11}(t)]}{dt} + \frac{F(t) - F_{11}(t)}{R}$$

$$C_p s F_{11}(s) = C_2 s [F(s) - F_{11}(s)] + \frac{F(s) - F_{11}(s)}{R}$$

$$C_p s F_{11}(s) = C_2 s F_{11}(s) + \frac{F(s)}{R} - \frac{F_{11}(s)}{R}$$

$$\left(C_p s + C_2 s + \frac{1}{R}\right) F_{11}(s) = \left(C_2 s + \frac{1}{R}\right) F(s)$$

$$R(C_p + C_2)s + 1 \; F_{11}(s) = (C_2 R s + 1) F(s)$$

$$F_{11}(s) = \frac{\frac{C_2 R s + 1}{R}}{\frac{R(C_p + C_2)s + 1}{R}} F(s) = \frac{C_2 R s + 1}{R(C_p + C_2)s + 1} F(s)$$

Circuito 2 Apagando $F(t)$

$$-aF(t) = RX(t) + \frac{1}{C_p+C_1}\int X(t)\,dt$$

$$F_{s2} = \frac{1}{C_p+C_1}\int X(t)\,dt$$

$$-aF(s) = RX(s) + \frac{X(s)}{(C_p+C_1)s}$$

$$F_{s1}(s) = \frac{X(s)}{(C_p+C_1)s}$$

$$F_{s2}(s) = \frac{X(s)}{-a}\left(\frac{R(C_p+C_1)s+1}{(C_p+C_1)s}\right)$$

$$\frac{F_{s2}}{F(s)} = \frac{\frac{X(s)}{(C_p+C_1)s}}{\frac{-A}{a}\,\frac{R(C_p+C_1)s+1}{(C_p+C_1)s}} = \frac{-a}{R(C_p+C_1)s+1}$$

$$F_2(s) = \frac{-aF(s)}{R(C_p+C_1)s+1}$$

Función de transferencia

$$F(s) = F_{s1}(s) + F_{s2}(s) = \frac{1}{(C_p+C_1)s}F(s) - \frac{aF(s)}{R(C_p+C_1)s+1}$$

$$F(s) = \frac{(C_pRs+1) - a}{R(C_p+C_1)s+1}\qquad \frac{F_{s2}}{F(s)}$$